

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MECHANICAL PROPERTIES OF FLUIDS

LECTURE- 4 PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1 Revision ✓
* viscosity
- 2 Stokes Law
- 3 Terminal Velocity
- 4 Surface Tension ✓
- 5 Some Phenomenon which are Explain on the Basic of Surface Tension

RECAP FROM 1ST LECTURE

Fluid

liquid

gas

Density ρ

$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Atmospheric pressure P_0

✓ Variation of pressure with depth

Repeat



$$P_B = P_A + \rho gh$$

✓ Pascal Law

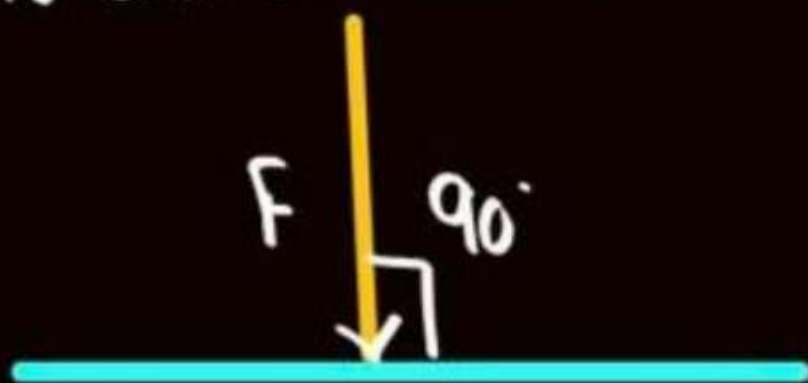
Recap-2

① Application pascal law

- Hydraulic Brake ✓
- Hydraulic lift

② Thrust

- Any perpendicular force is called thrust

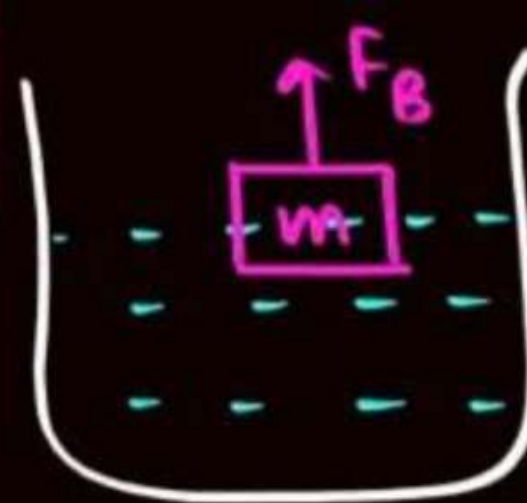


Buoyant force



Archimedes principle

$$F_B = \rho V g$$



Flow

Streamline flow

$$Re < 2000$$

Equation of Continuity

$$a_1 v_1 = a_2 v_2$$

v_c

turbulent flow

$$Re \gg \gg 2000$$

Recep - ③

Bernoullies Theorm

- law of conservation of Energy

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

Application of B-Theorm

① ✓ Spray / Atomizer

② ✓ Magnus effect

③ ✓ lift of aeroplane

Viscosity

lecture no. 4

①



It is the internal resistance of a liquid which oppose motion of layer of liquid



$A \rightarrow$ Area

$\frac{dv}{dx} \rightarrow$ velocity gradient

Maths

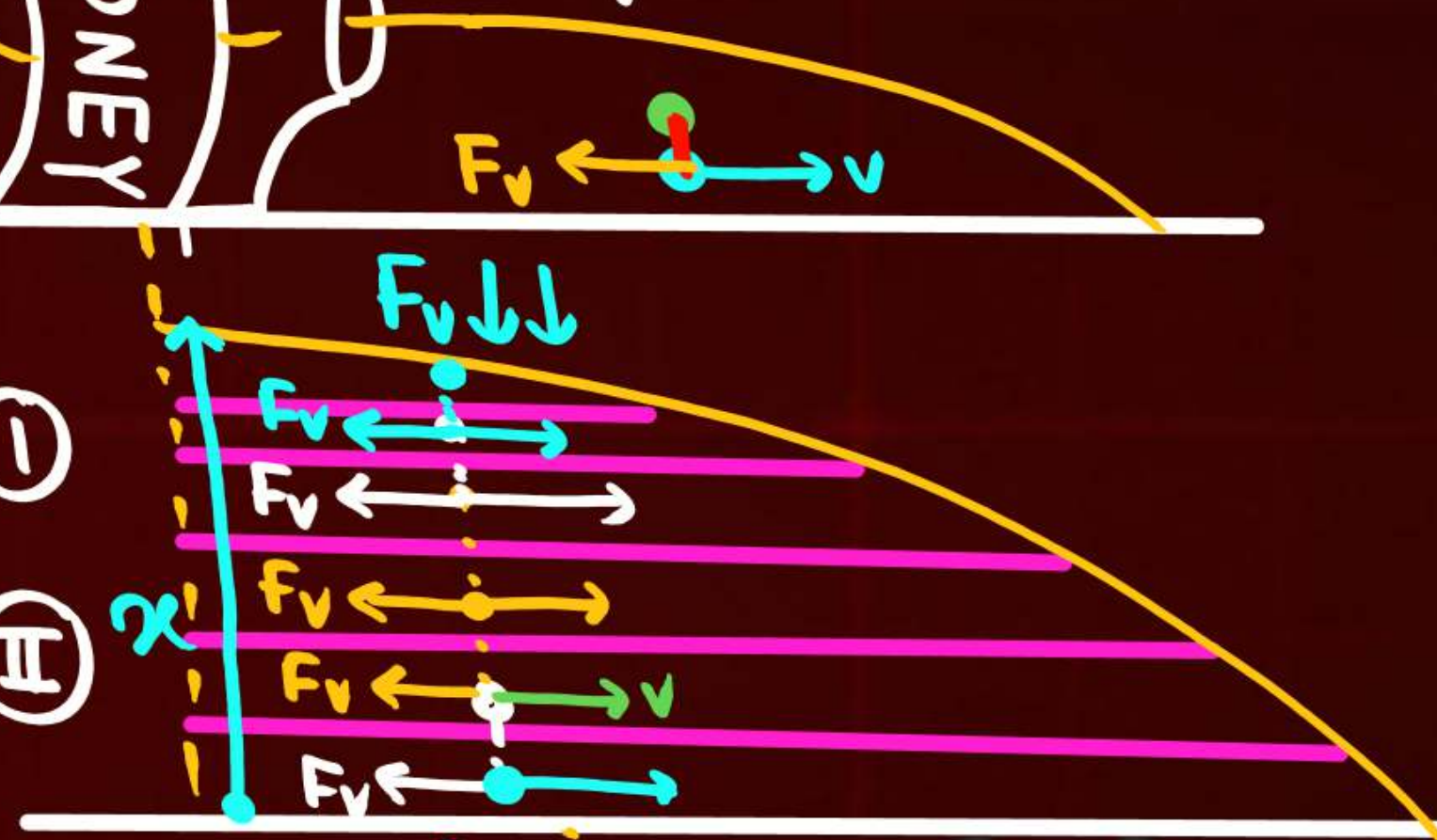
① $\uparrow F_v \propto A \uparrow$

② $\downarrow F_v \propto \frac{dv}{dx} \uparrow$

on combining ① and ②

$$F \propto A \frac{dv}{dx}$$

Viscosity $\propto \frac{F}{A \frac{dv}{dx}}$



$F_v \uparrow$

shazia takli



$$F_v \propto A \frac{dv}{dx}$$

$$F_v = -\eta A \frac{dv}{dx} \quad F_v \rightarrow \text{oppose the motion}$$

$\eta \Rightarrow$ coefficient of viscosity

S.I unit of η ✓

$$F_v = -\eta A \frac{dv}{dx}$$

$$N = \eta \, m^2 \frac{m}{s}$$

$$N = \eta \frac{m^2}{s}$$

$$\boxed{\frac{Ns}{m^2} = \eta}$$

Pascal Sec = η Neet-2023

②

$$\text{Pressure} = \frac{F}{A} = \frac{N}{m^2}$$

= Pascal

C.G.S unit of η

③

$$\underline{F_v} = -\eta A \underline{\frac{dv}{dx}}$$

$$v = \frac{\text{cm}}{\text{sec}}$$

$$\text{dynes} = \eta \text{ cm}^2 \frac{\cancel{\text{cm}}}{\text{sec } \cancel{\text{cm}}}$$

$$\text{dynes} = \eta \frac{\text{cm}^2}{\text{sec}}$$

$$\frac{\text{dynes sec}}{\text{cm}^2} = \eta$$

$$\text{Poise} = \eta$$

Dimensional Formula of η

2024 JKBOSE

④

$$F = \eta A \frac{dv}{dx}$$

$$MLT^{-2} = \eta L^2 \cancel{L} T^{-1}$$

$$MLT^{-2} = \eta L^2 \cancel{L} T^{-1}$$

$$\frac{M \cancel{L} T^{-2}}{\cancel{L}^2 T^{-1}} = \eta$$

$$ML^{-1} T^{-2} T^{+1} = \eta$$

$$F \rightarrow MLT^{-2}$$

$$A \rightarrow L^2$$

$$dv \rightarrow LT^{-1}$$

$$dx \rightarrow L$$

$$F_v = 6\pi\eta r v$$

$$\eta = ML^{-1} T^{-2+1}$$

$$\eta = ML^{-1} T^{-1}$$

$$\eta = ML^{-1} T^{-1}$$

Factor Effect the Viscosity

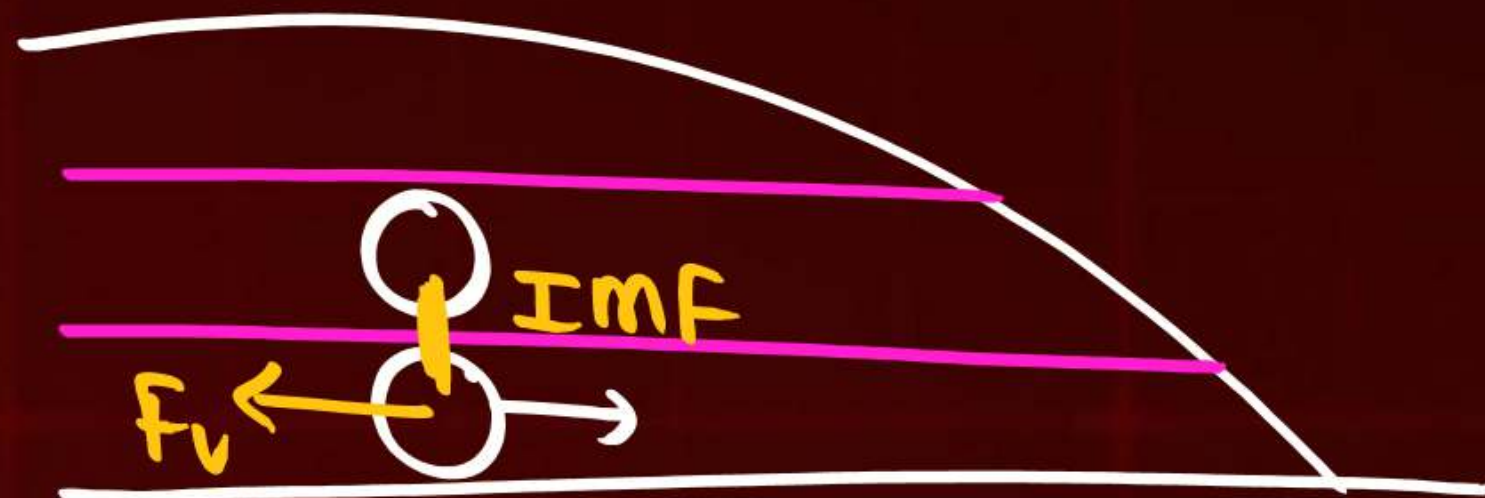
① Intermolecular Force

The inter molecular Force is directly proportional to viscos force (viscosity)

↓↓ $F_v \propto IMF$ ↓↓

② Temperature

Temperature is inversely proportional to viscos force



Stokes Law

$$F = ma$$

$$= 2 \times 2^2$$

$$= 4$$

$$6\pi = 6 \times 3.14$$

It give the viscous force on spherical

Body $F = 6\pi\eta r v$

Formula/magnitude of viscous force acting on spherical Body when it move through liquid

$F_v \propto v$ — ① velocity

$F_v \propto r$ — ② radius

$F_v \propto \eta$ — ③ coefficient of viscosity

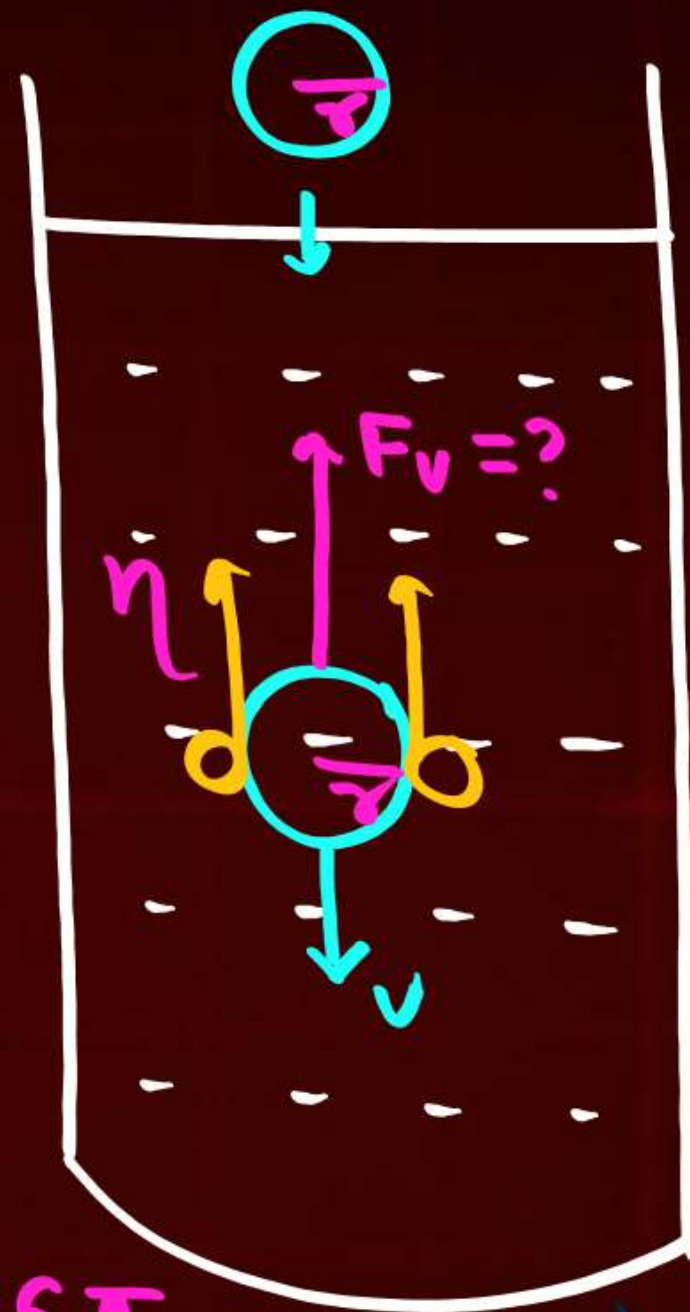
on combining ③ equation

$$F_v \propto v r \eta$$

$$F_v = K v r \eta \quad K = 6\pi$$

$$F_v = 6\pi v r \eta$$

$$F_v = 6\pi\eta r v$$



⑥



Can You Solve It?

JKBOSE - 2021, 2016, 2019, 2023



If viscous force ^{F_v} acting on spherical body moving through a liquid depend on velocity of Ball, radius of Ball and coefficient of viscosity.
Derive stroke law with the help of dimensional Analysis. ^{η}

$$F_v = 6\pi \eta r v$$

$$T = 2\pi \sqrt{l/g}$$

Sol $F_v \propto \eta^a \tau^b v^c$

$F_v \propto \eta^a \tau^b v^c$ — ①

write down all dimensional formula

$F_v = MLT^{-2}$ $\eta = ML^{-1}T^{-1}$ $\tau = L$

$v = LT^{-1}$

$MLT^{-2} \propto (ML^{-1}T^{-1})^a (L)^b (LT^{-1})^c$

$MLT^{-2} \propto M^a L^{-a} T^{-a} L^b L^c T^{-c}$

$MLT^{-2} \propto M^a L^{-a+b+c} T^{-a-c}$

$MLT^{-2} \propto M^a L^{-a+b+c} T^{-a-c}$

$MLT^{-2} \propto M^a L^{-a+b+c} T^{-a-c}$ ⑧

on comparing powers

$a=1$

$-2 = -a - c$

$-2 = -1 - c$

$c = -1 + 2$

$c = 2 - 1 = 1$

$1 = -a + b + c$

$1 = -1 + b + 1$

$1 = b$

$$\begin{aligned}\text{Put } a &= 1 \\ b &= 1 \\ c &= 1\end{aligned}$$

in equation 1st

$$F_v \propto \eta^a \tau^b v^c$$

$$F_v \propto \eta^1 \tau^1 v^1$$

$$F_v \propto \eta \tau v$$

$$F_v \propto \eta \tau v$$

⑨

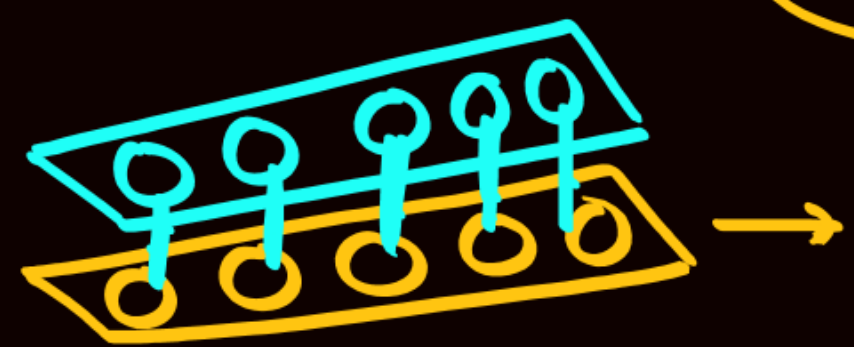
$$F_v = k \eta \tau v$$

$$k = 6\pi$$

$$F_v = 6\pi \eta \tau v$$

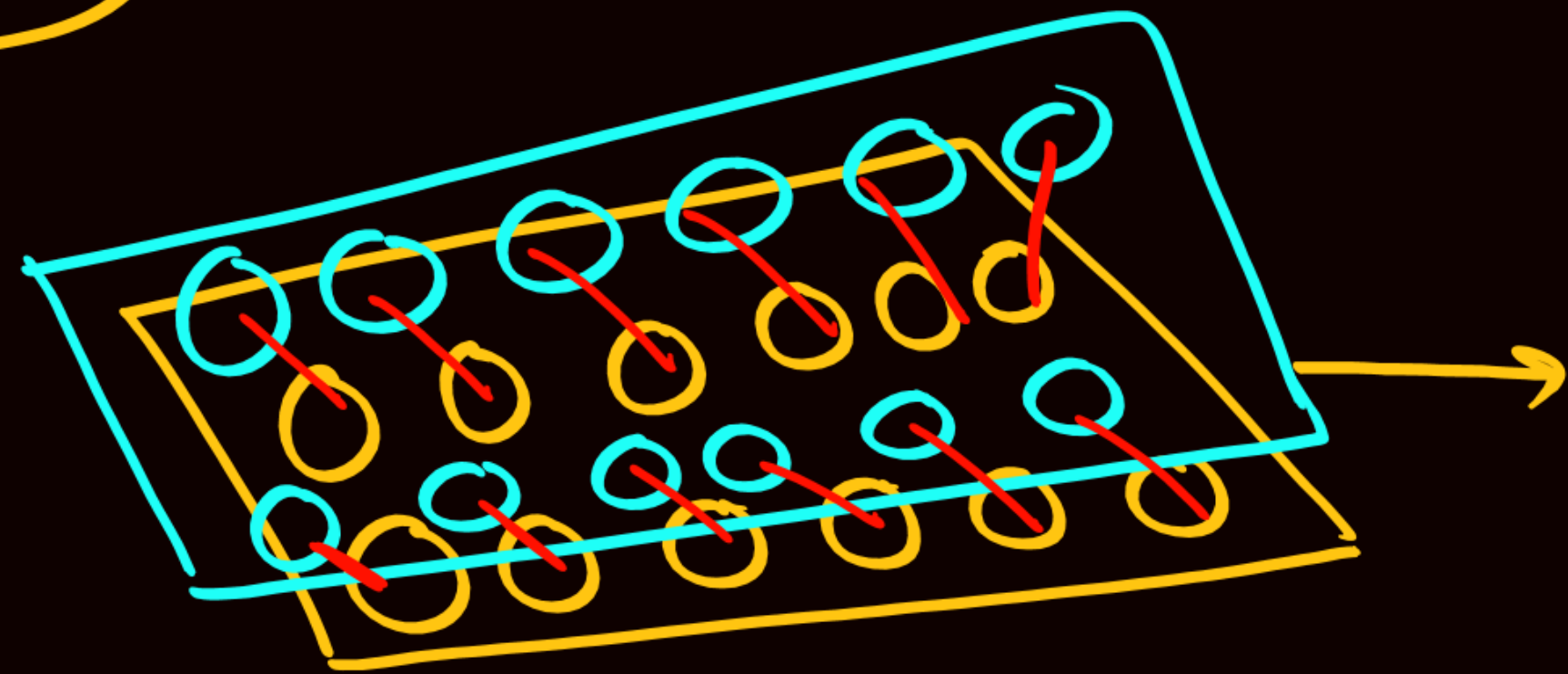
$\frac{dv}{dx}$ 12th

Rough



Imf = 5
 $f_v = 5$ A↓↓↓

$\frac{dv}{dx}$ → gradient
 $a = \frac{dv}{\textcircled{dt}}$ → rate



Imf = 12 A↑↑↑
 $f_v = 12$

$$F_v = 6\pi\eta r v$$

$$2 \times 3 \times 2 = 12$$

$$F_v = 6\pi\eta r v$$

$$3 \times 2 \times 2 = 12$$

Terminal Velocity

H/W



Can You Solve It?

JKBOSE - 2021, 2016, 2019, 2023



7

H/W If ^{F_v} viscous force acting on spherical body moving through a liquid depend on velocity of Ball, radius of Ball and coefficient of viscosity.
Try Derive stroke law with the help of dimensional Analysis. ^{η}

$$F_v = 6\pi \eta r v$$

$$T = 2\pi \sqrt{l/g}$$

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



Join Our Official Telegram & Whatsapp Channel



JKBOSE Wallah
Class 9th & 10th



JKBOSE Wallah



Scan the QR to Join

Thank
You

